

SLAYTON, MN, USA

WMO: 720368

Lat: 43.987N

Lon: 95.783W

Elev: 1623

StdP: 13.85

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 54924

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.5	-7.2	-17.6	2.2	-9.3	-14.4	2.7	-6.6	32.6	18.1	29.9	14.5	10.0	330	0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	88.9	71.1	86.1	70.1	83.3	68.4	75.7	83.8	73.8	81.6	71.8	79.4	13.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.3	131.4	80.6	71.4	123.0	78.7	69.4	114.8	76.9	40.4	84.2	38.5	81.7	36.6	79.4	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
28.6	25.9	23.5	DB	-16.3	93.2	4.4	2.6	-19.5	95.0	-22.1	96.5	-24.5	98.0	-27.7	99.8	
			WB	-16.8	78.2	4.3	2.1	-19.8	79.8	-22.4	81.0	-24.8	82.2	-27.9	83.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.9	15.8	17.8	31.9	44.1	57.2	67.4	71.4	68.5	61.0	47.9	34.2	19.7
	DBStd	22.13	12.94	12.07	12.74	10.45	9.17	6.33	5.67	5.42	8.18	9.63	11.26	11.85
	HDD50	4388	1061	901	574	230	34	0	0	0	9	155	484	940
	HDD65	7899	1526	1321	1028	628	276	46	12	26	171	535	925	1405
	CDD50	2524	0	0	12	53	257	523	664	574	340	92	9	0
	CDD65	557	0	0	0	1	34	118	211	134	52	7	0	0
	CDH74	5031	0	0	3	33	412	1082	1893	1036	496	76	0	0
	CDH80	1430	0	0	0	4	133	317	585	253	125	13	0	0
Wind	WSAvg	10.3	12.2	11.8	11.4	12.3	11.5	9.1	7.2	6.7	8.2	10.3	11.7	10.7
Precipitation	PrecAvg	29.3	0.6	0.9	1.5	3.2	3.3	4.7	3.5	3.1	3.6	2.4	1.4	1.0
	PrecMax	46.0	1.8	1.9	3.7	7.6	7.0	11.9	9.4	6.9	10.7	5.6	4.3	2.1
	PrecMin	21.5	0.0	0.2	0.0	1.4	0.7	2.1	0.9	0.3	0.9	0.4	0.0	0.2
	PrecStd	6.0	0.5	0.5	1.1	1.5	1.7	2.1	1.9	2.2	2.6	1.6	1.4	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.6	54.1	71.2	79.1	90.3	92.0	92.2	89.5	88.3	81.7	68.5	49.7
		MCWB	38.2	49.6	53.8	58.9	63.7	70.7	75.2	73.6	68.0	63.0	54.0	44.4
	2%	DB	38.8	41.6	62.2	72.0	83.4	86.9	89.2	86.0	84.0	75.0	60.3	42.4
		MCWB	34.7	37.3	52.4	54.8	62.8	68.2	74.0	72.2	67.9	59.2	49.8	37.7
	5%	DB	35.9	37.2	55.0	66.3	78.7	83.6	86.2	82.5	80.2	69.6	55.0	37.2
		MCWB	33.4	34.1	49.0	51.2	60.8	66.2	72.1	70.2	65.4	56.1	45.8	34.4
	10%	DB	33.1	33.8	49.4	61.0	73.1	80.5	83.1	80.1	76.1	63.7	49.8	34.3
		MCWB	31.7	31.8	45.1	49.2	58.6	64.4	70.2	68.6	63.6	53.0	43.3	32.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.6	49.0	60.0	60.6	69.3	75.8	79.5	77.4	74.9	67.8	57.3	45.7
		MCDB	43.7	54.2	66.6	71.2	81.9	84.1	87.5	85.5	81.9	76.6	62.8	48.3
	2%	WB	35.2	38.5	53.8	57.2	66.6	72.4	76.5	74.8	71.3	61.9	50.7	38.0
		MCDB	37.8	41.7	58.7	67.9	76.6	81.1	84.4	82.3	78.2	71.0	58.3	41.2
	5%	WB	33.5	34.4	49.0	54.3	63.7	69.4	74.6	72.8	68.4	57.2	47.0	34.9
		MCDB	35.4	36.7	54.8	62.8	74.0	78.8	83.0	79.8	76.3	66.9	54.3	37.2
	10%	WB	31.8	32.0	44.4	51.2	61.2	67.5	72.7	70.8	66.0	53.7	43.3	32.5
		MCDB	33.2	33.5	49.6	59.0	69.9	76.7	80.7	77.2	73.4	61.5	49.8	34.0
Mean Daily Temperature Range	5% DB	MDBR	15.8	16.0	15.8	19.2	20.2	19.3	20.0	19.5	21.8	19.7	17.5	14.4
		MCDBR	15.8	18.7	25.6	29.3	27.5	24.4	21.8	21.9	25.4	27.9	24.4	16.8
		MCWBR	12.9	14.5	16.2	15.4	12.3	10.7	10.4	10.8	11.5	14.6	14.6	13.3
	5% WB	MCDBR	15.7	17.4	23.3	24.7	22.0	20.6	20.0	19.5	21.5	23.6	22.0	15.2
		MCWBR	13.3	13.9	15.4	15.1	11.4	11.0	10.8	11.1	12.1	14.1	14.0	12.3
Clear-Sky Solar Irradiance	taub		0.259	0.274	0.306	0.361	0.379	0.394	0.404	0.405	0.360	0.319	0.284	0.265
	taud		2.455	2.431	2.442	2.323	2.342	2.359	2.319	2.287	2.390	2.492	2.515	2.457
	Ebn at Noon		281	295	297	286	283	278	274	269	274	274	269	267
	Edh at Noon		22	27	31	38	39	38	39	39	33	26	21	20
All-Sky Solar Radiation	RadAvg		595	933	1218	1514	1749	1950	2096	1730	1371	915	615	459
	RadStd		61	94	112	143	128	133	76	134	99	116	47	44

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page